P8131 Spring 2026 Homework #7 Solution

Due on April 13 at 11:59pm

1. In a pitch study, we are interested in the relationship between pitch and politeness. There are two levels politeness (a formal register: pol, and an informal register: inf). On top of that, we also have an additional fixed effect, gender. Each subject was tested on several scenarios (e.g., asking a peer for a favor (informal condition) or asking a professor for a favor (formal condition)). The pitch measurements are typically correlated for the same subject and in the same scenario.

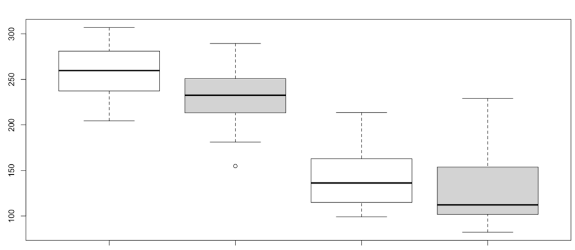
   (a) Exploratory analysis: provide boxplots to show the relation between gender/attitude and pitch (ignoring different scenarios).

   (b) Fit a mixed effects model with random intercepts for different subjects (gender and attitude being the fixed effects). What is the covariance matrix for a subject Y_i? What is the covariance matrix for the estimates of fixed effects (Hint: 3×3 matrix for intercept, gender and attitude)? What are the BLUPs for subject-specific intercepts? What are the residuals?

   (c) Fit a mixed effects model with intercepts for different subjects (gender, attitude and their interaction being the fixed effects). Use likelihood ratio test to compare this model with the model in part (b) to determine whether the interaction term is significantly associated with pitch.

   (d) Write out the mixed effects model with random intercepts for both subjects and scenarios (gender and attitude being the fixed effects). Fit the model using lmer in the lme4 package. Write out the covariance matrix for a subject Y_i. What is the interpretation of the coefficient for the fixed effect term attitude?

Solution:



(a) Among males and females, the median line is lower for the polite than for the informal condition. There may be a bit more overlap between the two politeness categories for males than for females.

(b)

$$y_{ij} = \beta_0 + \beta_1 genderM_i + \beta_2 attitudePOL_{ij} + \alpha_{0i} + \epsilon_{ij}$$

where, $i = 1, \ldots, 6$, $j = 1, \ldots, 14$.

$$\hat{var}(y_i) = \begin{pmatrix} \hat{var}(\alpha_0) + \hat{var}(\epsilon) & \ldots & \hat{var}(\alpha_0) \\ \vdots & \ddots & \vdots \\ \hat{var}(\alpha_0) & \ldots & \hat{var}(\alpha_0) + \hat{var}(\epsilon) \end{pmatrix}_{14\times 14} = \begin{pmatrix} 1446 & \ldots & 598 \\ \vdots & \ddots & \vdots \\ 598 & \ldots & 1446 \end{pmatrix}$$

$$\hat{var}(\boldsymbol{\beta}) = \begin{pmatrix} 230 & -219 & -20 \\ -219 & 439 & 0 \\ -20 & 0 & 40 \end{pmatrix}$$

$$BLUP = \begin{pmatrix} F1 : -13.58 \\ F2 : 10.17 \\ F3 : 3.41 \\ M3 : 27.96 \\ M4 : 4.74 \\ M7 : -32.70 \end{pmatrix}$$

Residuals:

1	2	3	4	5	6
-10.1086926	-38.9110735	61.6913074	16.2889265	-19.5086926	43.4889265
7	8	9	10	11	12
27.3913074	33.3889265	8.4913074	8.9889265	-42.2086926	-12.7110735
13	14	15	16	17	18
-26.9110735	-68.6086926	-10.6898326	-23.0922136	-3.5898326	-9.3922136
19	20	21	22	23	24
26.6101674	5.6077864	35.0101674	46.4077864	-7.7898326	-7.8922136
25	26	27	28	29	30
-13.8898326	18.4077864	4.0077864	-54.8898326	-22.2262298	-29.3286108
31	32	33	34	35	36
96.0737702	-38.0286108	-20.7262298	60.6713892	60.4737702	9.9713892
37	38	39	40	41	42
-31.1262298	-26.0286108	-22.9262298	-16.7286108	-6.9286108	-6.4262298
43	44	45	46	47	48
-9.3872915	-16.3896725	-13.2872915	-11.1896725	-9.5872915	-5.2896725
49	50	51	52	53	54
1.6127085	4.5103275	-1.7872915	-12.5896725	13.3127085	-7.2896725
55	56	57	58	59	60
8.9103275	12.1127085	-14.4550462	-35.8574271	-0.8550462	-7.4574271
61	62	63	64	65	66
42.2449538	34.6425729	-3.9550462	29.0425729	30.5449538	27.0425729
67	68	69	70	71	72
-39.1550462	-41.2574271	13.8425729	-19.9550462	-2.3471929	12.6504261
73	74	75	76	77	78
-13.7471929	23.5504261	4.0528071	9.9504261	51.3528071	14.7504261
79	80	81	82	83	84
4.5528071	-19.6495739	-9.4471929	-18.1495739	-15.0495739	-2.8471929

(c)

$$y_{ij} = \beta_0 + \beta_1 genderM_i + \beta_2 attitudePOL_{ij} + \beta_3 genderM_i * attitudePOL_{ij} + \alpha_{0i} + \epsilon_{ij}$$

$H_0 : \beta_3 = 0$ vs $H_1 : \beta_3 \neq 0$, test statistics = 1.39, P-value is 0.24, we fail to reject H_0 at 5% level of significance. The interaction term is not significantly associated with pitch.

(d)

$$y_{ijk} = \beta_0 + \beta_1 genderM_i + \beta_2 attitudePOL_{ij} + \alpha_{0i} + \gamma_{0k} + \epsilon_{ijk}$$

where, i indicates subject i,j indicates attitude, $j = 1, 2$. k indicates scenario, $k = 1, \ldots, 7$.

```
Random effects:
 Groups   Name        Variance Std.Dev.
 scenario (Intercept) 224.5    14.98
 subject  (Intercept) 613.2    24.76
 Residual             637.8    25.25
Number of obs: 84, groups:  scenario, 7; subject, 6

Fixed effects:
            Estimate Std. Error t value
(Intercept)  256.987     16.101  15.961
genderM     -108.798     20.956  -5.192
attitudepol  -20.002      5.511  -3.630

Correlation of Fixed Effects:
            (Intr) gendrM
genderM     -0.651
attitudepol -0.171  0.000
```

You can see that scenario has much less variability than subject. The covariance structure for a subject is

$$\begin{aligned}
\hat{var}(y_{ijk}) &= \hat{var}(\alpha_{0i}) + \hat{var}(\gamma_{0k}) + 2\hat{cov}(\alpha_{0i}, \gamma_{0k}) + \hat{var}(\epsilon_{ijk}) \\
&= 1475.5 \\
\hat{cov}(y_{ijk}, y_{ijm}) &= \hat{var}(\alpha_{0i}) + \hat{cov}(\alpha_{0i}, \gamma_{0k}) + \hat{cov}(\alpha_{0i}, \gamma_{0m}) + \hat{cov}(\gamma_{0m}, \gamma_{0k}) \\
&= 613.2 \\
\hat{cov}(y_{ijk}, y_{inm}) &= \hat{var}(\alpha_{0i}) \\
&= 613.2 \\
\hat{cov}(y_{ijk}, y_{ink}) &= \hat{var}(\alpha_{0i}) + 2\hat{cov}(\alpha_{0i}, \gamma_{0k}) + \hat{var}(\gamma_{0k}) \\
&= 837.7
\end{aligned}$$

$$\hat{var}(y_i) = \begin{pmatrix} 1476 & \ldots & 613 & 838 & \ldots & 613 \\ \vdots & 1476 & \vdots & \vdots & 838 & \vdots \\ 613 & \ldots & 1476 & 613 & \ldots & 838 \\ 838 & \ldots & 613 & 1476 & \ldots & 613 \\ \vdots & 838 & \vdots & \vdots & 1476 & \vdots \\ 613 & \ldots & 838 & 613 & \ldots & 1476 \end{pmatrix}$$

The interpretation of the fixed effect attitude is on average polite decreases 20 Hz pitch comparing to informal.

Appendix

```
rm(list = ls())
library(lme4)

data <- read.csv('/politeness_data.csv')
summary(data)
boxplot(frequency ~ attitude*gender, col=c("white","lightgray"),data= data)

m1 <- lmer(frequency ~ gender + attitude + (1|subject), REML = F, data = data)
summary(m1)
round(ranef(m1)$subject,2)
residuals(m1)

m2 <- lmer(frequency ~ gender + attitude + gender*attitude+ (1|subject),
REML = F, data = data)
summary(m2)
ranef(m2)
TS1 <- deviance(m1) - deviance(m2)
p1 <- 1-pchisq(TS1, 1)

m3 <- lmer(frequency ~ gender + attitude + (1+attitude|subject), data = data)
summary(m3)
ranef(m3)

m4 <- lmer(frequency ~ gender + attitude + (1|subject)+(1|scenario), data = data)
summary(m4)
ranef(m4)
```